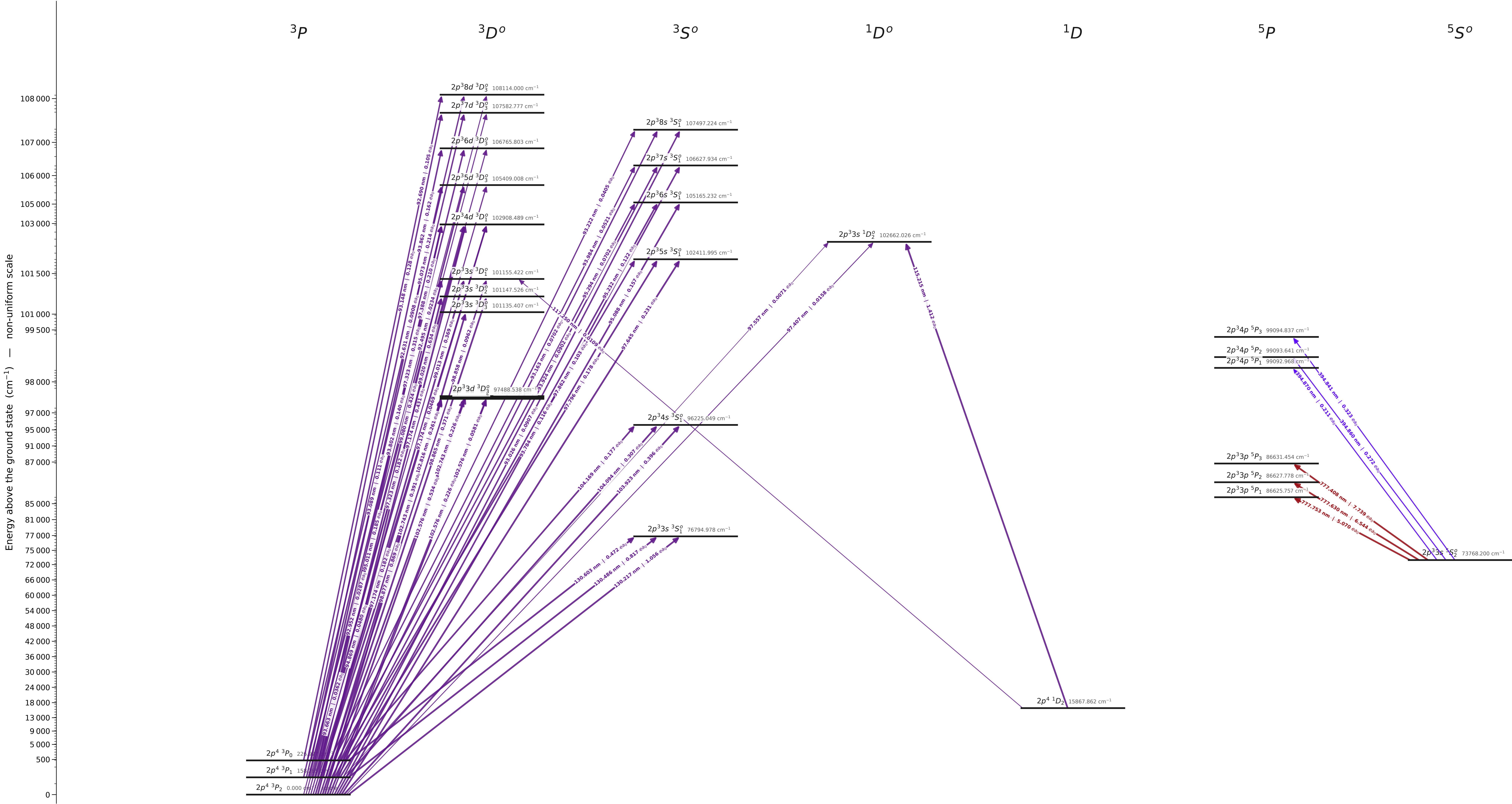
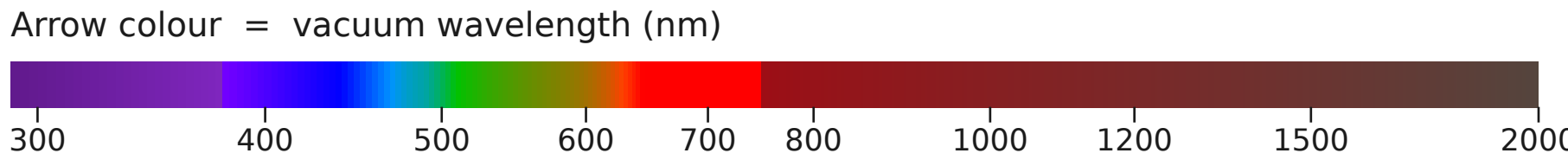


Oxygen

O (O I). Strongest classified transitions below 2 μm (NIST): 31 levels, 58 lines.

O⁺ ionisation limit 109837.02 cm⁻¹ = 13.61805 eV (91.044 nm)

Arrow colour = vacuum wavelength | reduced dipole matrix element $\langle U||er||J' \rangle$ | in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U er J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$2p^4\ ^3P_2 - 2p^34d\ ^3D_2^o$	97.1738	-	3085.1154	0.431	5.850e+07	-	-	
$2p^4\ ^3P_1 - 2p^34d\ ^3D_2^o$	97.3234	-	3080.3728	0.315	4.370e+07	-	-	
$2p^4\ ^3P_2 - 2p^33s\ ^3D_2^o$	98.8655	-	3032.3265	0.371	5.770e+07	-	-	
$2p^4\ ^3P_2 - 2p^33s\ ^3D_2^o$	98.8773	-	3031.9632	0.869	2.260e+08	-	-	
$2p^4\ ^3P_1 - 2p^33s\ ^3D_2^o$	99.0127	-	3027.8186	0.369	9.470e+07	-	-	
$2p^4\ ^3P_1 - 2p^33s\ ^3D_2^o$	99.0204	-	3027.5819	0.634	1.680e+08	-	-	
$2p^4\ ^3P_0 - 2p^33s\ ^3D_2^o$	99.0801	-	3025.7587	0.424	1.250e+08	-	-	
$2p^4\ ^3P_2 - 2p^33d\ ^3D_2^o$	102.5762	-	2922.6328	0.534	7.660e+07	-	-	
$2p^4\ ^3P_1 - 2p^33d\ ^3D_2^o$	102.7431	-	2917.8855	0.391	5.710e+07	-	-	
$2p^4\ ^3P_0 - 2p^33d\ ^3D_2^o$	102.8157	-	2915.8234	0.261	4.220e+07	-	-	
$2p^4\ ^3P_2 - 2p^34s\ ^3S_1^o$	103.9230	-	2884.7544	0.396	9.430e+07	-	-	
$2p^4\ ^3P_1 - 2p^34s\ ^3S_1^o$	104.0943	-	2880.0097	0.307	5.640e+07	-	-	
$2p^4\ ^1D_2 - 2p^33s\ ^1D_2^o$	115.2151	-	2602.0236	1.412	5.280e+08	-	-	
$2p^4\ ^3P_2 - 2p^33s\ ^3S_1^o$	130.2168	-	2302.2555	1.056	3.410e+08	-	-	
$2p^4\ ^3P_1 - 2p^33s\ ^3S_1^o$	130.4858	-	2297.5109	0.817	2.030e+08	-	-	
$2p^4\ ^3P_0 - 2p^33s\ ^3S_1^o$	130.6029	-	2295.4509	0.472	6.760e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).